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Non-temporal neural networks for predicting degradation trends of key wind-turbine gearbox components☆

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ABSTRACT

Deep-learning models, which are widely used to predict degradation trends, are effective for automatic extraction of hidden features from monitoring data and complete learning of the evolutionary patterns of component performance. However, wind turbine monitoring systems often yield discontinuous data owing to short-term sensor abnormalities or unstable data transmission links, leading to difficulties in training existing deep-learning models and low prediction accuracy for the degradation trends of key gearbox components. This study proposes a nontemporal neural network (NTNNet) with multitask training for predicting the degradation trends of key wind-turbine components using discontinuous data. Multiple support and query sets are constructed by intercepting short time-series samples, and multitask training is performed to overcome model training difficulties in the case of discontinuous samples. A gated recurrent unit is constructed using the meta-learning framework, and parameters are updated through multiple recursions to enhance the characterization ability for key-component performance. The accuracy and superiority of the proposed method are validated using two datasets and compared to existing methods. The proposed method achieves degradation trend prediction accuracies of 4.61, 4.73, 4.61, and 3.05 for the four components of wind turbine gearboxes, respectively, meetingthe practical application requirements for the health assessment of wind turbine gearboxes.

1. Introduction

With the rapid development of the wind-turbine industry, a large operation and maintenance market is emerging, with maintenance being a focus [1–3]. The gearbox is the core component of a wind-turbine transmission system and, although its failure frequency is low, its repair and maintenance are extremely difficult and time-consuming, causing the longest downtime. Therefore, gearbox failures seriously affect power generation and cause significant economic losses [4–6]. If the degradation trends of the key gearbox components can be accurately predicted, faults can be detected in time and corresponding measures can be taken. Hence, wind-turbine reliability can be improved and service lifetimes can be prolonged [7,8]. Therefore, research on predicting the degradation trends of key gearbox components is of great significance for reducing wind-turbine operation and maintenance costs.

Deep learning models are effective in automatically extracting hidden features from monitoring data and can be sufficient to learn the performance evolution patterns of key components. Therefore, they are often used to predict degradation trends of key components in wind turbine gearboxes [9–11]. However, the prediction accuracy of existing deep-learning methods regarding these degradation trends relies heavily on the temporal features of the data. Under non-continuous sample conditions, primarily due to poor training-data timing, deep neural networks typically exhibit poor generalization ability, leading to

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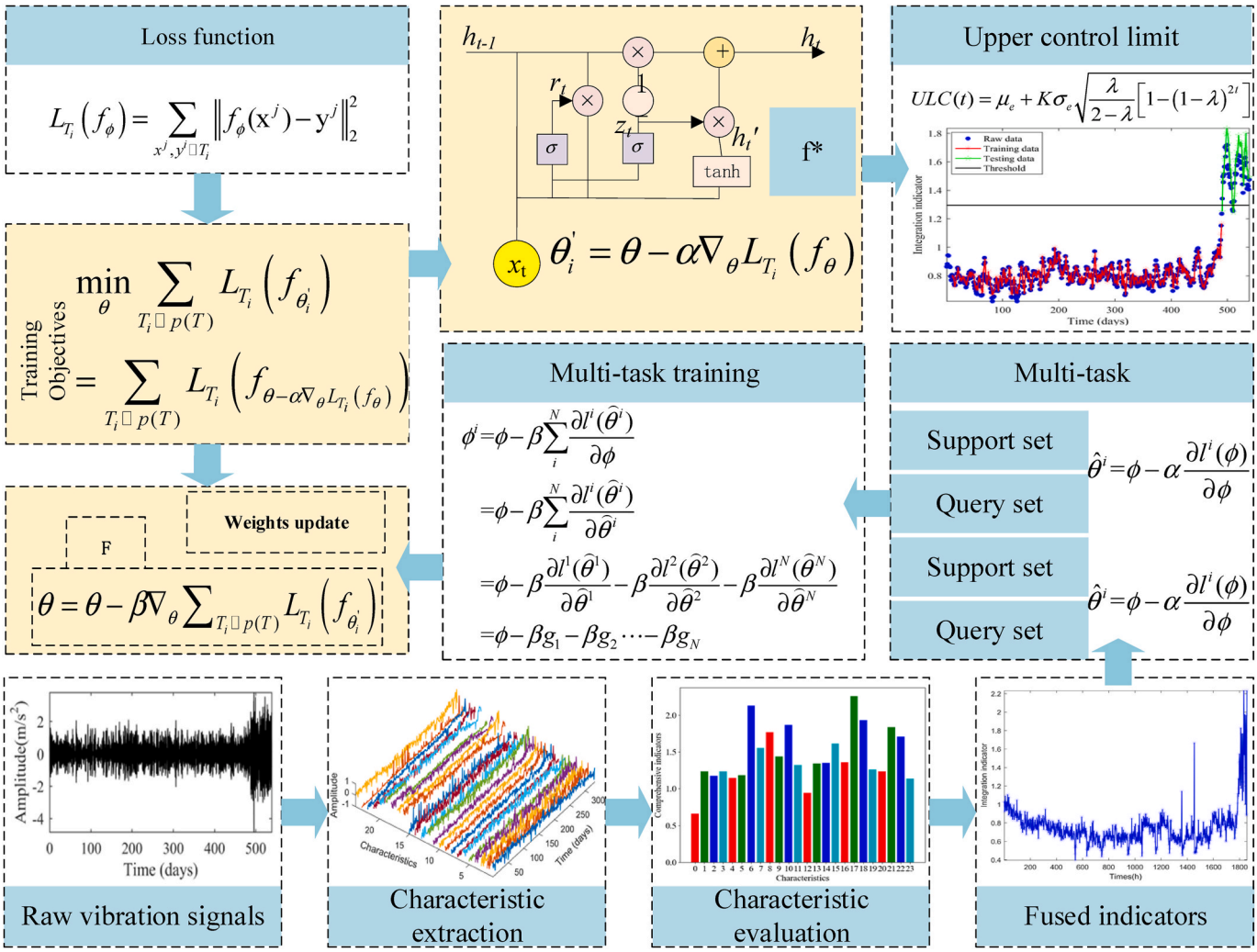


Fig. 1. Prediction model of key wind-turbine gearbox components based on NTNNet.

inaccurate degradation trend predictions. Various factors cause discontinuity in training samples for predicting the degradation trends of key wind-turbine gearbox components, including short-term sensor faults and a lack of long-term data collection systems operable during wind-turbine gearbox operation [12]. Therefore, improving prediction accuracy for the degradation trends of key wind-turbine gearbox components under non-continuous sample conditions is a challenge.

To date, various deep neural networks have been applied to predict the degradation trends of key wind-turbine gearbox components. For example, Xiang et al. [13] proposed a concise self-adapting deep learning network consisting of a multi-branch 1D involution neural network, multi-head gated recurrent unit (GRU), and Mish activation function, which demonstrated superior performance in predicting the remaining useful lives (RULs) of wind-turbine gearbox bearings. Bai [14] proposed the two-phase-based deep neural network, which enables rolling-bearing health monitoring through the use of logarithmic squared envelope-based diversity entropy and rolling-bearing RUL prediction via an initial RUL prediction, with uncertainty quantification being performed via a Bayesian neural network. Cao [15] proposed the combination of parallel GRUs with a dual-stage attention mechanism, and incorporated uncertainty quantification for improved wind-turbine bearing RUL prediction accuracy. Note that dual-stage attention enhances degradation extraction, a parallel structure enhances accuracy, and a nonparametric approach provides probabilistic results. Yan et al. [16] extracted frequency-domain characteristics from vibration signals to construct a gear degradation-trend index and used long short-term memory networks (LSTMs) to quantitatively evaluate gear health statuses. Yu et al. [17] designed a novel bidirectional LSTM (Bi-LSTM), which achieved gear degradation-trend prediction for a small number of non-full-life samples. The bootstrap method was used to obtain the RUL confidence interval and a Bi-GRU network was used to predict the gear degradation trends. However, challenges remain: adjustment of the initial model parameters is difficult, and the model generalization performance is poor when discontinuous samples are provided. Therefore, the prediction accuracy of the degradation trends of key wind-turbine gearbox components requires further improvement.

Meta-learning [18–20] can adaptively adjust model initialization parameters through multitask learning, enabling the model to quickly converge and overcome the problem of reduced prediction accuracy owing to poor generalization performance under discontinuous sample conditions. This approach can be applied to predict the degradation trends of key gearbox components, such as the gears and bearings [21]. For example, Yang et al. [21,22] proposed a novel MetaDESK model based on meta-learning with a deep sparse kernel network. The MetaDESK model first trains a sparse kernel using a variational posterior in a data-driven manner and then transfers this kernel to a new few-shot RUL task via meta-knowledge. Pan et al. [23] proposed a multi-head attention network integrated with adaptive meta-transfer learning for RUL (Remaining Useful Life) prediction. This approach combines a convolution-based branch with an attention-based branch. The multi-head attention network is specifically designed to improve the accuracy of RUL prediction for bearings during steady operational

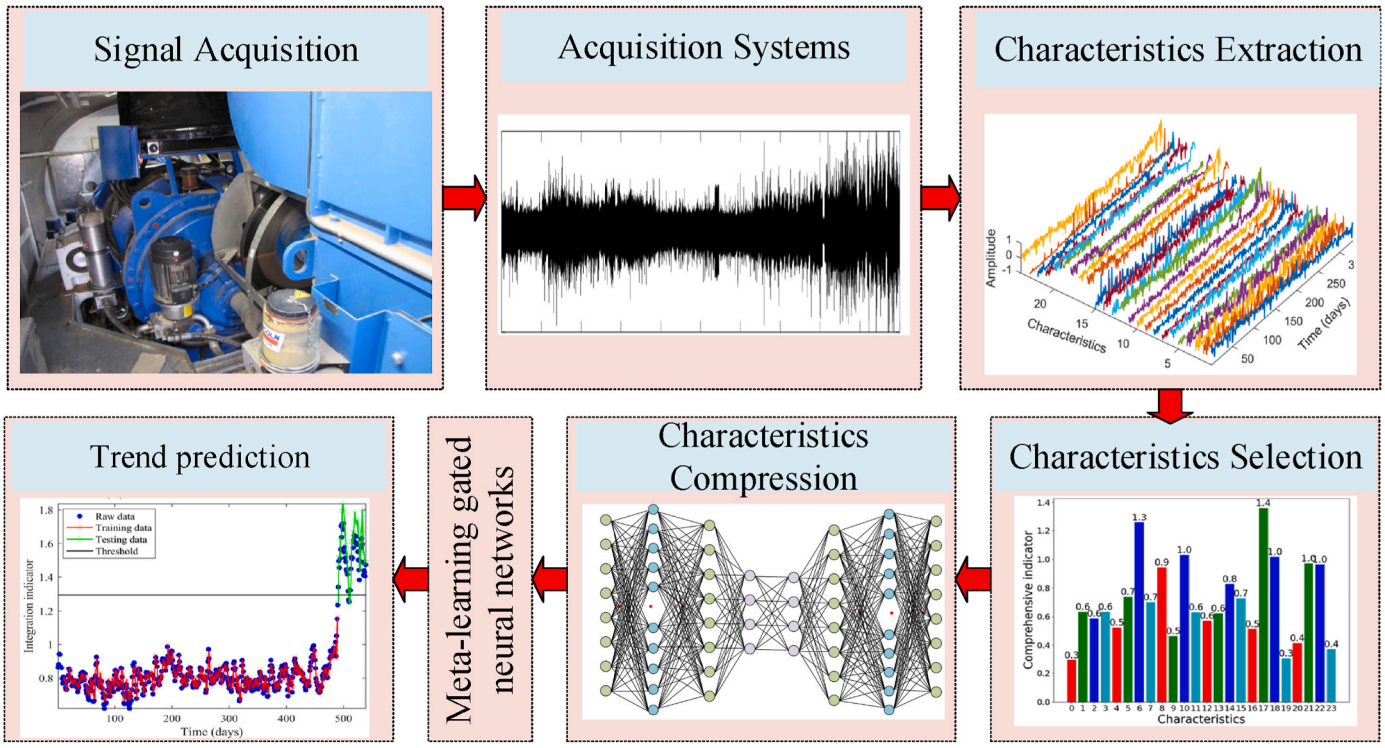


Fig. 2. Degradation-trend prediction process for key wind-turbine gearbox components.

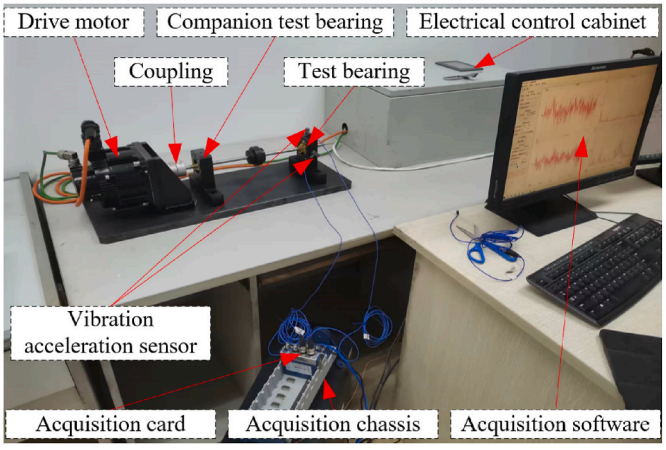
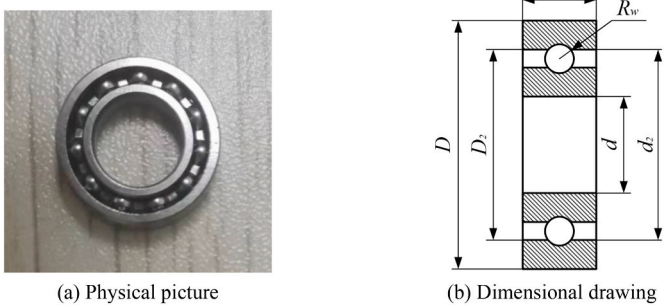


Fig. 3. Bearing fatigue-test rig.



(a) Physical picture (b) Dimensional drawing

Fig. 4. Tested deep-groove ball bearing.

Table 1
Dimensional parameters of tested bearings.

Basic Size/(mm)						Rolling element
d	D	B	d_2	D_2	D_w	Z
17	26	5	19.83	23.17	2.4	16

stages. Zhao et al. [24] proposed a fault diagnosis framework based on unsupervised meta-learning and contrastive learning. This framework separates a task into two subsets, which are then fed into the inner and outer loops, respectively, to achieve fault diagnosis in rotating machinery. However, the degradation patterns of components in wind turbine gearboxes are highly complex. Under conditions of

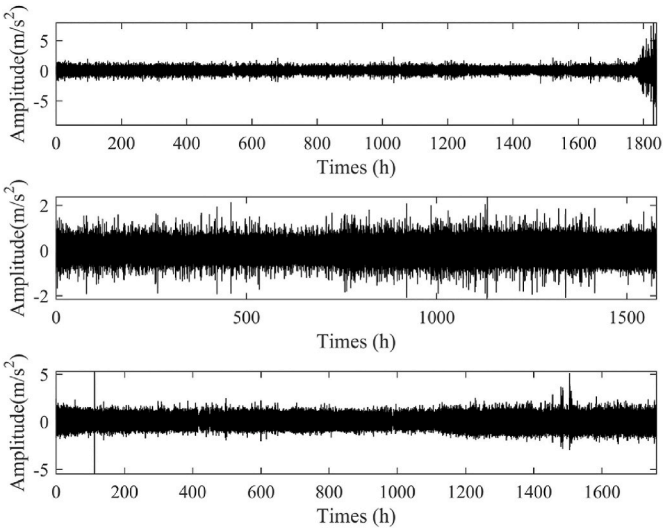


Fig. 5. Raw vibration signals from bearing fatigue tests.

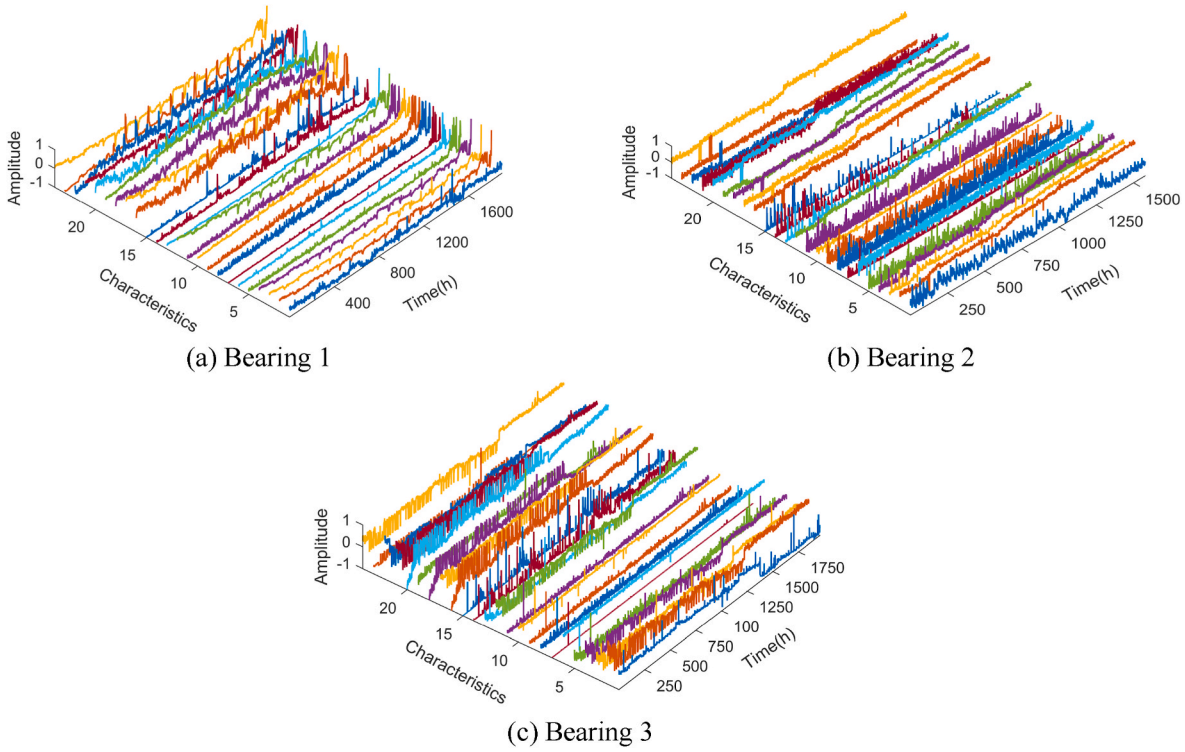


Fig. 6. Time- and frequency-domain characteristics of bearing vibration signals.

discontinuous data, a single meta-learning network struggles to accurately predict the degradation trends of these components.

To address these problems, this study proposes a method for predicting the degradation trends of key wind-turbine gearbox components that is based on a non-temporal neural network (NTNNet). In this approach, multitask learning is used to train the NTNNet, fully extract features under discontinuous sample conditions, adaptively adjust the model initialization parameters, strengthen the NTNNet representation ability for degradation mechanisms, and improve the prediction accuracy of the degradation trends of key wind-turbine gearbox components. The main contributions of this study are as follows.

(1) Raw vibration signals of key wind-turbine gearbox components are processed using characteristic extraction, selection, and fusion to generate indicators that act as inputs for the proposed NTNNet.
(2) The problem of model training on discontinuous samples is addressed by sampling short time-series data, constructing multiple support and query sets, and utilizing the multitask learning training method.
(3) A GRU is constructed under the meta-learning framework and the parameters are updated through multiple recursions to enhance the representation ability for the key-component performance and to accurately predict degradation trends.

The remainder of this paper is organized as follows. Section 2 introduces the construction method used to obtain the integrated indicators for the key wind-turbine gearbox components. Section 3 describes the construction of the proposed NTNNet for predicting the degradation trends of wind turbine gearboxes. Subsequently, the proposed method is validated using a bearing dataset collected from a self-built bearing fatigue test rig, as described in Section 4. Section 5 applies the model to wind-turbine gearbox health management, achieving degradation-trend prediction of key components such as bearings and gears. Finally, Section 6 summarizes the study and discusses future research directions. pages.

2. Integrated-indicator construction

To fully understand and model the degradation trends of key components in a wind-turbine gearbox, it is essential to construct fused indicators that can serve as inputs for prediction models. The process involves extracting relevant time-domain and frequency-domain features from the vibration signals of the gearbox, evaluating these features to select those that exhibit monotonic behavior and strong trendability, using a Deep Auto-Encoder (DAE) model to fuse the selected features into a more comprehensive representation, and finally, constructing fused indicators that effectively reflect the degradation trends of the key components. By following these steps, the fused indicators will enable more accurate and reliable predictions, facilitating timely maintenance and reducing the risk of unexpected failures.

2.1. Characteristic extraction and selection

During the degradation of key wind-turbine gearbox components such as gears and bearings, both the time- and frequency-domain characteristics of the vibration signals exhibit change patterns. To analyze the degradation patterns and predict the degradation trends of the key components effectively, the method proposed in this study extracts 11 time- and 13 frequency-domain vibration-signal characteristics [25] to construct integrated indicators. To select effective characterization indicators, the monotonicity and trendability are used to evaluate the characteristic quality [26]. The monotonicity is calculated as follows:

$$Mon = \left| \frac{No.ofd/dx > 0}{T-1} - \frac{No.ofd/dx < 0}{T-1} \right|, \tag{1}$$

where Mon represents the monotonicity of the extracted vibration-signal characteristic; $No.ofd/dx > 0$ and $No.ofd/dx < 0$ are the numbers of positive and negative derivatives of the characteristic sequence, respectively; and T is the length of the characteristic sequence. The value of Mon ranges from 0 to 1, and the closer Mon is to 1, the better the

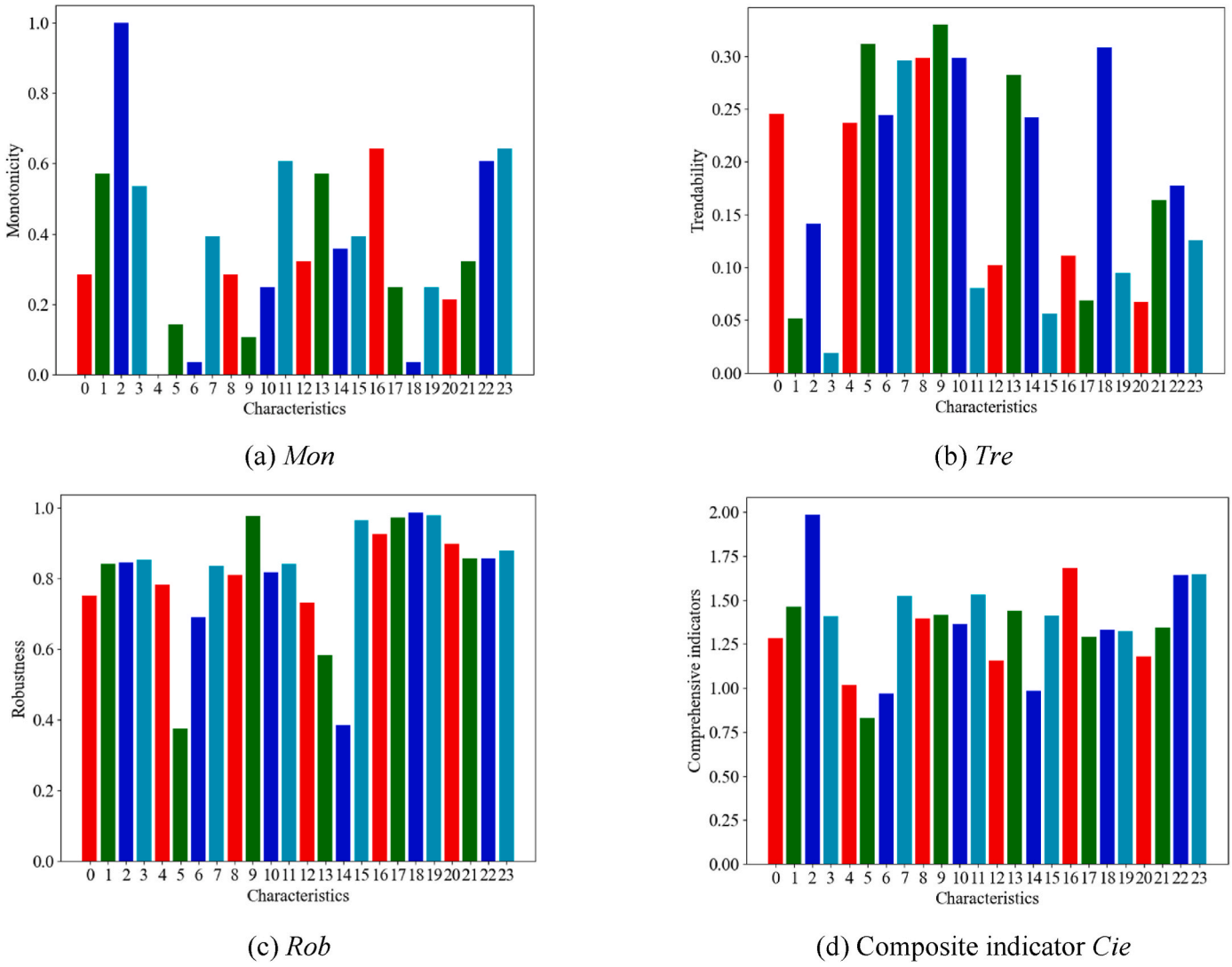


Fig. 7. Gear characteristic evaluation results.

monotonicity of the characteristic. The trendability is calculated as follows:

$$Tre = \frac{\left|K\left(\sum_{k=1}^{K}\widetilde{x}_k\widetilde{t}_k\right) - \left(\sum_{k=1}^{K}\widetilde{x}_k\right)\left(\sum_{k=1}^{K}\widetilde{t}_k\right)\right|}{\sqrt{\left[K\sum_{k=1}^{K}\widetilde{x}_k^2 - \left(\sum_{k=1}^{K}\widetilde{x}_k\right)^2\right]\left[K\sum_{k=1}^{K}\widetilde{t}_k^2 - \left(\sum_{k=1}^{K}\widetilde{t}_k\right)^2\right]}}, \tag{2}$$

where *Tre* represents the correlation between the degradation trend of the characteristic and time, $\{\widetilde{x}_k\}$ represents the characteristic extracted from the vibration signal, and $\{\widetilde{t}_k\}$ denotes the corresponding time series of the characteristic. A *Tre* value closer to 1 indicates stronger temporal correlation of the characteristic.

In addition, robustness is used to evaluate the proposed characteristics:

$$Rob = \frac{1}{T}\sum_{t=1}^{T}\exp\left(-\left|\frac{x_t - x_t^{Tv}}{x_t}\right|\right), \tag{3}$$

where x_t denotes the value of the feature curve at time t and x_t^{Tv} denotes the value of the curve at time t after exponential smoothing. *Rob* takes the value interval [0,1]; the higher the value of *Rob*, the better the robustness of the characteristic.

To comprehensively characterize the monotonicity, trendability, and robustness of the characteristics, a composite indicator is employed for the characteristic evaluation:

$$Cie = Mon + Tre + Rob. \tag{4}$$

where *Cie* represents the composite indicator of the characteristic, *Mon* represents the monotonicity, *Tre* represents the trendability, and *Rob* represents the robustness.

2.2. Characteristic integration

To better characterize the degradation patterns of key wind-turbine gearbox components, eight characteristics were selected and fused based on consideration of both *Mon* and *Tre*, in accordance with previous empirical studies [27,28]. In this study, a DAE network was primarily utilized for characteristic fusion. A DAE is primarily composed of multiple encoding and decoding layers. Its encoding layer can be represented as

$$y_i(x) = \sigma(W_i x + b_i), \tag{5}$$

where x is the input feature matrix; i represents the i-th encoding layer; W_i and b_i are the i-th layer weight and bias, respectively; $\sigma(\cdot)$ is the activation function; and y_i is the i-th layer output. After the features are encoded layer-by-layer, the decoding layer is used to decode the network. Here,

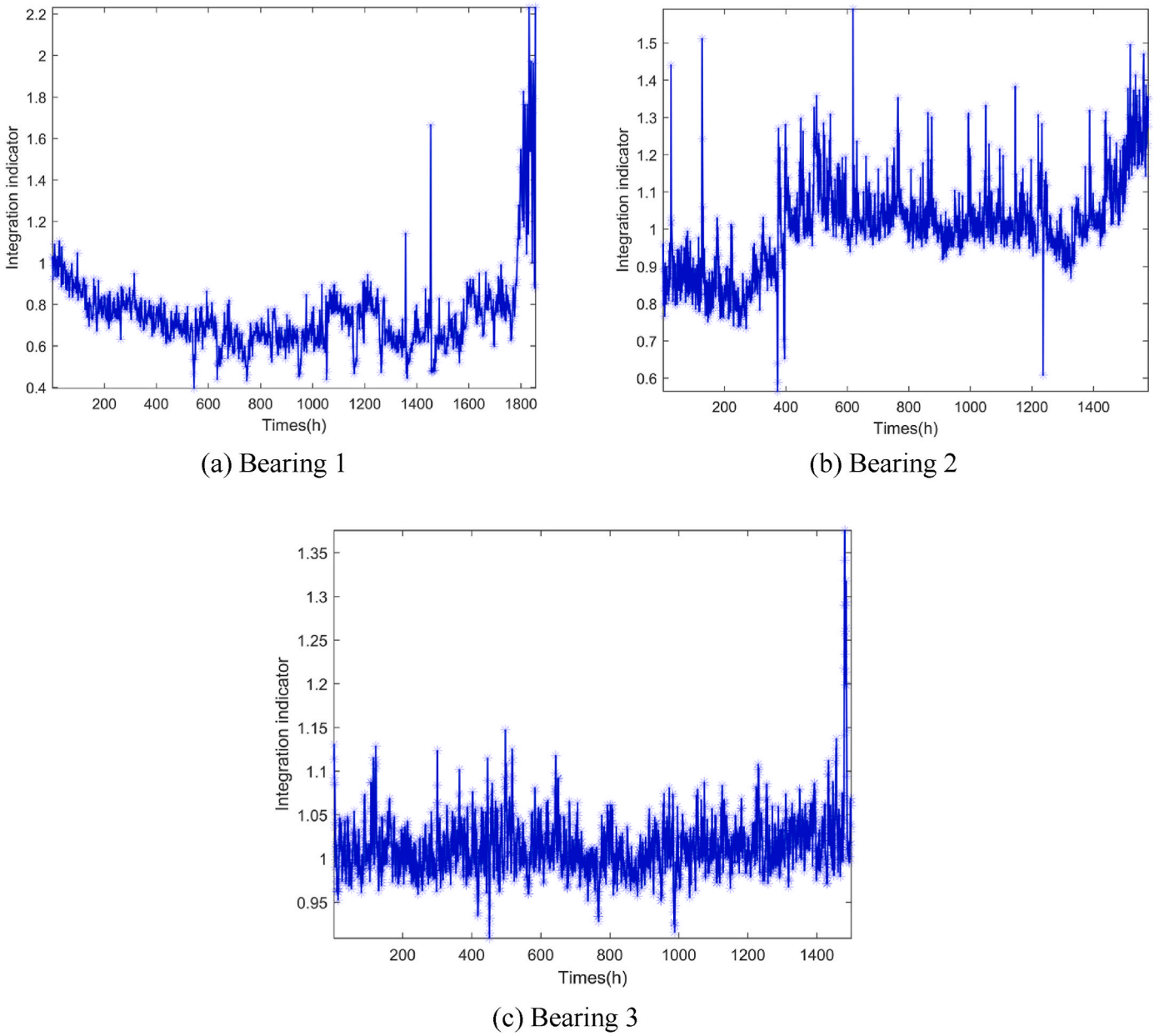


Fig. 8. Integrated indicator results.

Fig. 9. Photographs showing disassembly of tested bearing.

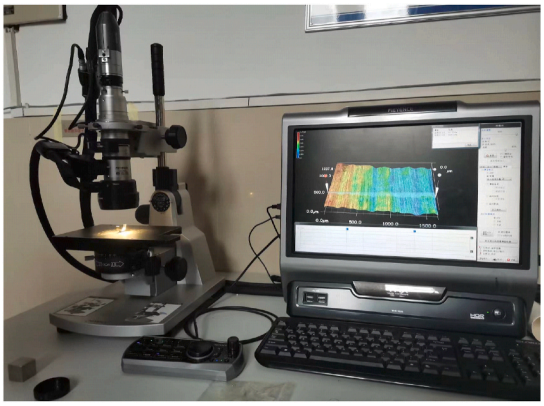


Fig. 10. 3D super depth-of-field measurement system.

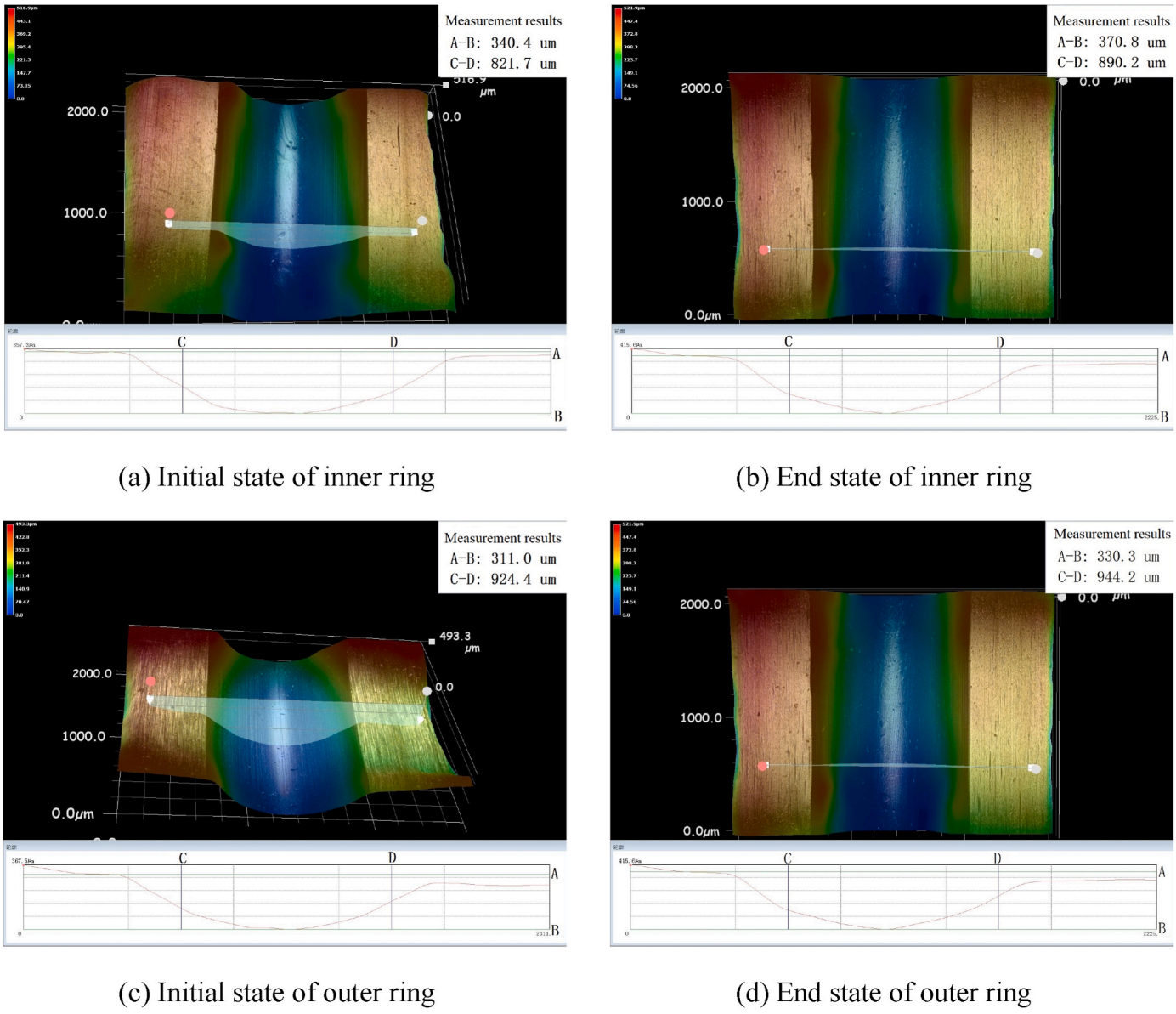


Fig. 11. Wear measurement results for Bearing 1.

$$x_i' = \sigma(W_i y_i + b_i), \tag{6}$$

where y_i and x_i' represent the input and output data of the i-th decoding layer, respectively. During the model training, the input and output reconstruction error ($RE = x' - x$) is used to optimize the model. The minimal RE indicates that the model has sufficiently extracted the input-data features. Finally, the output of the DAE-network middle layer is used as the final fusion indicator for the key wind-turbine gearbox components.

3. Proposed method

3.1. Proposed non-temporal neural networks

To address the issue of poor feature extraction in existing deep learning models under conditions of data discontinuity, which leads to low prediction accuracy of degradation trends in wind turbine gearbox components, an NTNNet with multi-task training is proposed to predict the degradation trends of key wind turbine gearbox components using discontinuous data. The proposed method is illustrated in Fig. 1.

As shown in Fig. 1, the proposed prediction method primarily comprises several stages: signal acquisition, feature extraction, feature evaluation, fusion indicator construction, and degradation trend prediction. First, the fusion indicator is structured as multi-task, with each task incorporating different support sets and query sets to reduce the impact of data discontinuity on the model. Subsequently, the various sub-tasks are input into a meta-learning framework to construct a gated recurrent unit, and parameters are recursively updated multiple times to enhance the feature extraction capability of the proposed model. Finally, the degradation trend prediction of critical components in wind turbine gearboxes is achieved by integrating the upper control limit (UCL).

The NTNNet degradation-trend prediction model is constructed, which includes multiple GRUs, with each unit defined as an f function. When the data of a task are input to f, the parameter θ of f is updated to θ_i'. The gradient updating method is

$$\theta_i' = \theta - \alpha \nabla_\theta L_{T_i}(f_\theta), \tag{7}$$

where α represents a model gradient-updating hyperparameter and $L_{T_i}(f_\theta)$ represents the loss of f that is learned from the support set of a task. Here, f is a gated recurrent neural network that includes update and

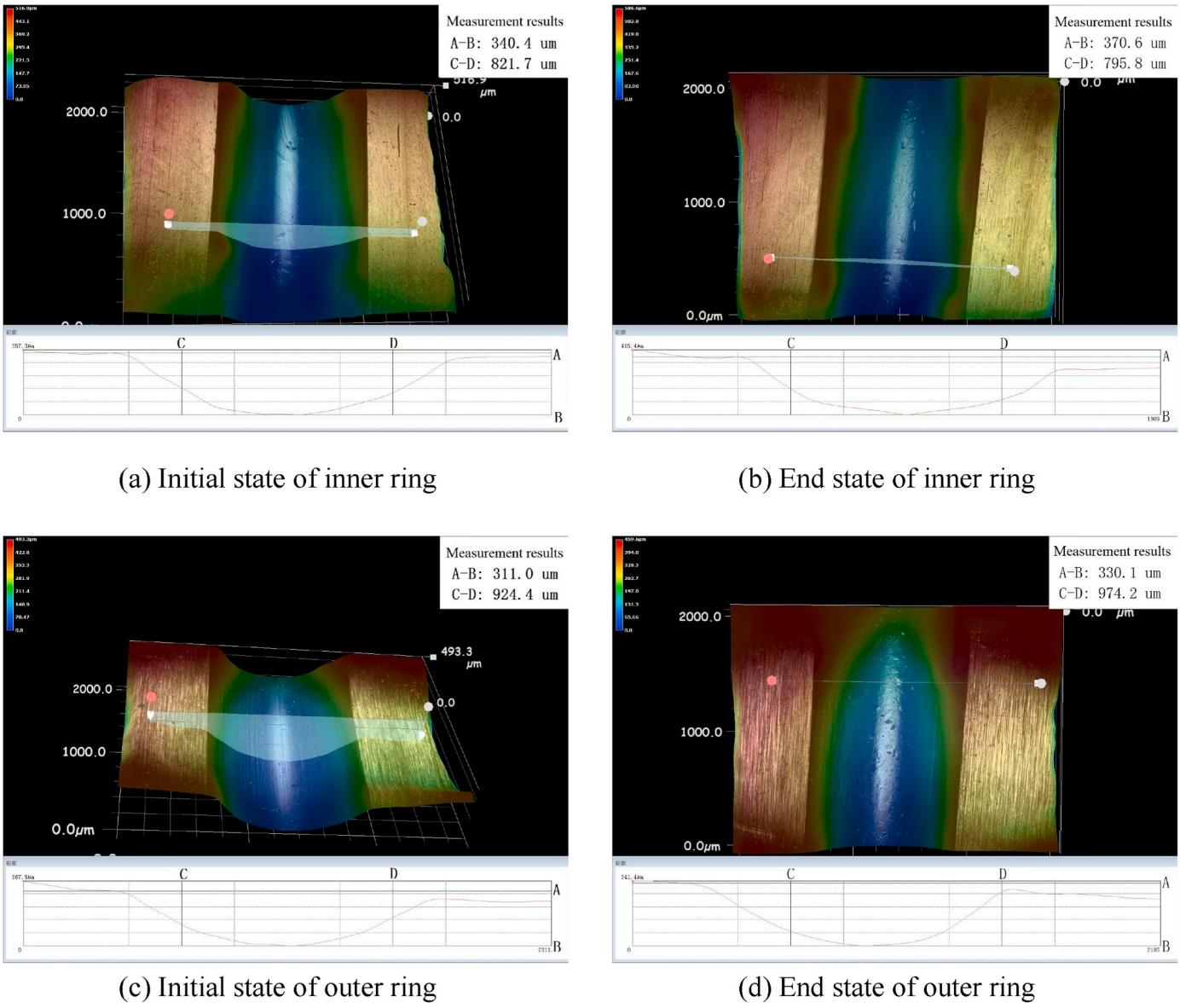


Fig. 12. Wear measurement results for Bearing 2.

reset gates z_t and r_t, respectively. Assuming that the input to f is the fused indicator x_t and the output data is the predicted value h_t, the input–output relationship can be expressed as

$$z_t = \sigma\left(W^{(z)}x_t + U^{(z)}h_{t-1}\right), \tag{8}$$

$$r_t = \sigma\left(W^{(r)}x_t + U^{(r)}h_{t-1}\right), \tag{9}$$

$$h_t' = \tanh(r_t \odot Uh_{t-1} + Wx_t), \tag{10}$$

$$h_t = (1 - z_t) \odot h_t' + z_t \odot h_{t-1}, \tag{11}$$

where x_t and h_t represent the input and output data, respectively; $\sigma(\cdot)$ denotes the sigmoid activation function; tanh is the hyperbolic tangent function; $\odot$ represents the element-wise product of matrices; $W^{(z)}$, $U^{(z)}$, $W^{(r)}$, $U^{(r)}$, U, and W are the trainable parameters; and h_t' denotes the sum of x_t and the previous hidden-layer state information h_{t-1}.

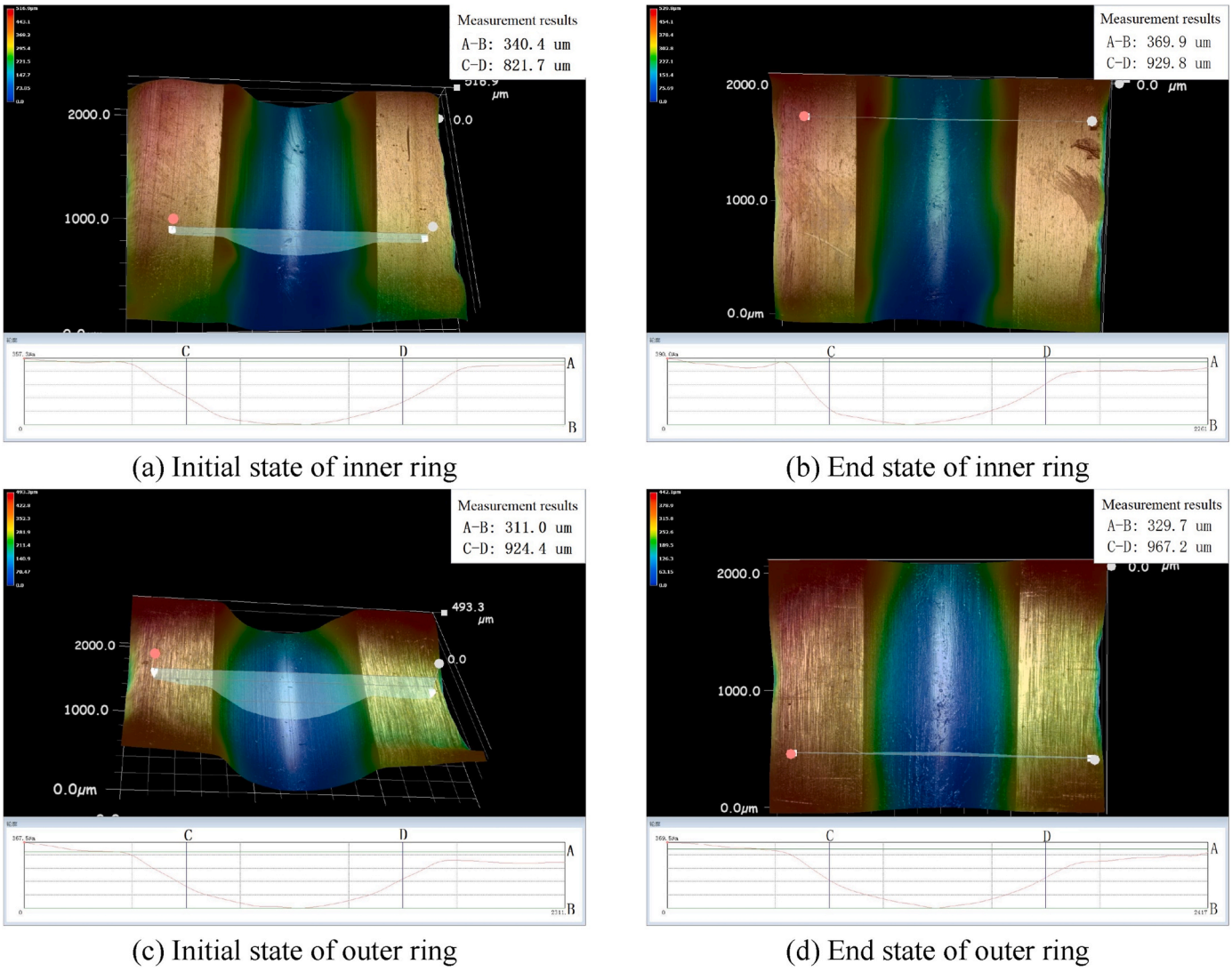
When the model learns the support set from the training data, the model parameters are trained and updated as follows:

$$\widehat{\theta}^i = \phi - \alpha\frac{\partial l^i(\phi)}{\partial \phi}, \tag{12}$$

where ϕ is the initial parameter and $l^i(\phi)$ is the loss of the support set in the i-th task during model training. The objective of learning multiple support sets is

$$\phi^* = \text{argmin}\left[\sum_i^N l^i\left(\widehat{\theta}^i\right)\right], \tag{13}$$

where ϕ^* represents the updated parameters after gradient descent and N is the number of training samples for a single task. The function argmin( ) indicates that the network parameters at this point are taken as ϕ^* when the total loss of all tasks reaches the minimum. After a set of tasks is learned, the model parameter update can be expressed as



(a) Initial state of inner ring (b) End state of inner ring

(c) Initial state of outer ring (d) End state of outer ring

Fig. 13. Wear measurement results for Bearing 3.

$$\begin{aligned}\phi^i &= \phi - \beta\frac{\partial L(\phi)}{\partial\phi}\\ &= \phi - \beta\sum_i^N \frac{\partial l^i\left(\widehat{\theta}^i\right)}{\partial\phi},\end{aligned} \tag{14}$$

where β is the meta-learning update step size, $\widehat{\theta}^i$ is the result of an inner update of ϕ, $l^i\left(\widehat{\theta}^i\right)$ is the query set loss for the i-th task, and $L(\phi)$ is the total loss of the model after all tasks are learned.

Following learning of the support set of a task in the network, the parameters are updated using the query set. As $\widehat{\theta}_j = \phi_j - \alpha\partial l(\phi)/\partial\phi_j$, the derivative of $\partial\widehat{\theta}_j$ with respect to $\partial\phi_i$ can be expressed as

$$\frac{\partial\widehat{\theta}_j}{\partial\phi_i} = \begin{cases}\alpha\dfrac{\partial l(\phi)}{\partial\phi_j\partial\phi_i} & i\neq j\\ 1-\alpha\dfrac{\partial l(\phi)}{\partial\phi_j\partial\phi_i} & i=j\end{cases}, \tag{15}$$

where i represents the i-th task and j represents the j-th training sample in a single task. When taking the derivative of a single task, higher-order derivatives can be ignored and a first-order approximation can be used. The derivative of a single task can be expressed as

$$\frac{\partial l\left(\widehat{\theta}\right)}{\partial\phi_i} = \sum_j^m \frac{\partial l\left(\widehat{\theta}\right)}{\partial l\left(\widehat{\theta}_j\right)}\frac{\partial\widehat{\theta}_j}{\widehat{\theta}\phi_i} \approx \frac{\partial l\left(\widehat{\theta}\right)}{\partial\widehat{\theta}_i}. \tag{16}$$

Because ϕ represents various network parameters, it can be represented by a vector. Therefore, the derivative of $l\left(\widehat{\theta}\right)$ with respect to ϕ can be expressed as

$$\frac{\partial l\left(\widehat{\theta}\right)}{\partial\phi} = \begin{bmatrix}\partial l\left(\widehat{\theta}\right)/\partial\phi_1\\ \partial l\left(\widehat{\theta}\right)/\partial\phi_2\\ \vdots\\ \partial l\left(\widehat{\theta}\right)/\partial\phi_i\end{bmatrix} \approx \begin{bmatrix}\dfrac{\partial l\left(\widehat{\theta}\right)}{\partial\widehat{\theta}_1}\\ \dfrac{\partial l\left(\widehat{\theta}\right)}{\partial\widehat{\theta}_2}\\ \vdots\\ \dfrac{\partial l\left(\widehat{\theta}\right)}{\partial\widehat{\theta}_i}\end{bmatrix} = \frac{\partial l\left(\widehat{\theta}\right)}{\partial\theta}. \tag{17}$$

Substituting Eq. (16) into Eq. (13) and simplifying yields

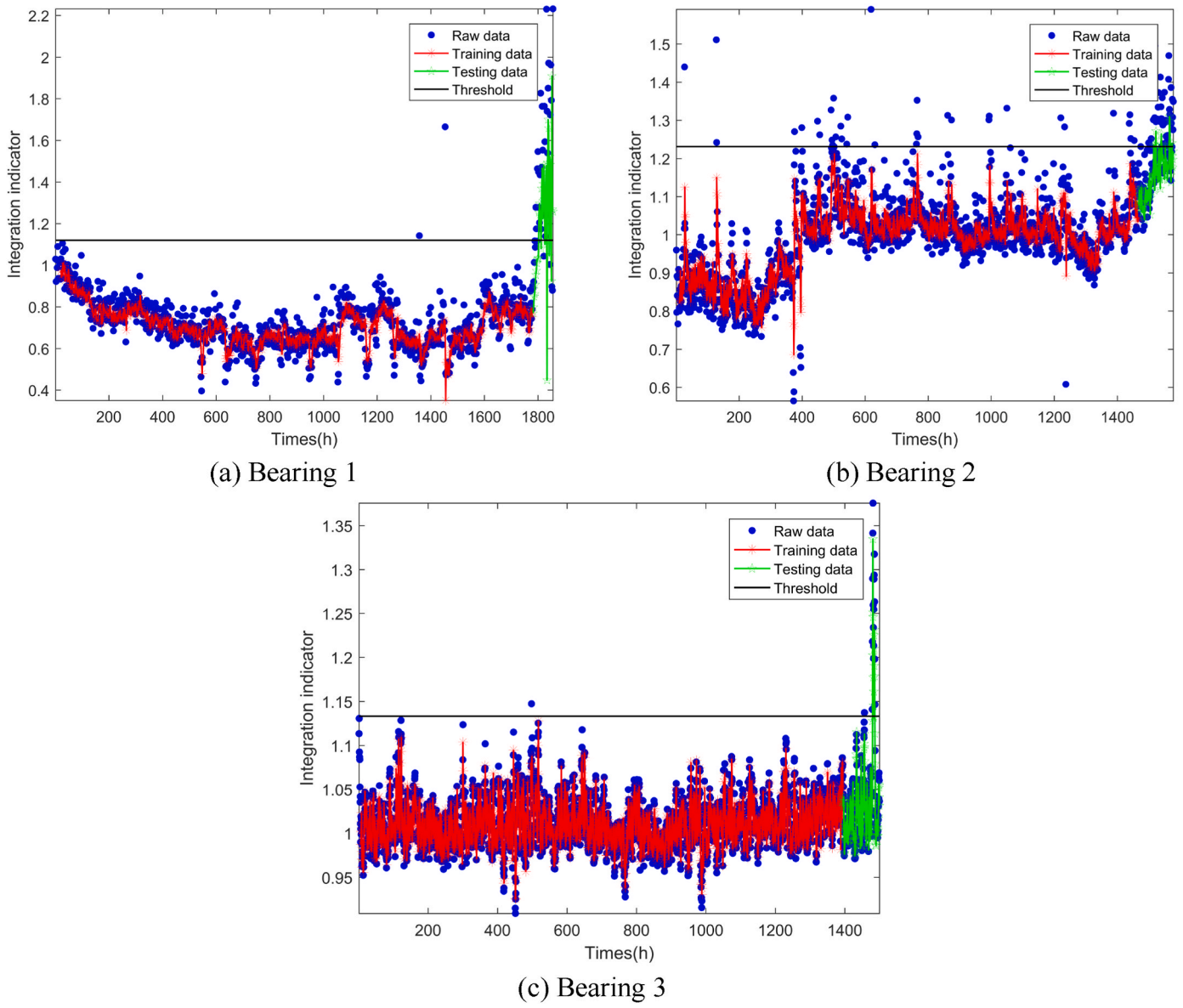


Fig. 14. Bearing degradation-trend prediction results.

$$\begin{aligned}\phi^i &= \phi - \beta\sum_i^N \frac{\partial l^i\left(\widehat{\theta}^i\right)}{\partial \phi} \\ &= \phi - \beta\sum_i^N \frac{\partial l^i\left(\widehat{\theta}^i\right)}{\partial \widehat{\theta}^i} \\ &= \phi - \beta\frac{\partial l^1\left(\widehat{\theta}^1\right)}{\partial \widehat{\theta}^1} - \beta\frac{\partial l^2\left(\widehat{\theta}^2\right)}{\partial \widehat{\theta}^2}\cdots - \beta\frac{\partial l^N\left(\widehat{\theta}^N\right)}{\partial \widehat{\theta}^N} \\ &= \phi - \beta g_1 - \beta g_2 \cdots - \beta g_N,\end{aligned} \tag{18}$$

where g_i denotes the gradient descent direction for the i-th task. Thus, after the model learns a set of tasks, the parameter update method for F can be simplified and expressed as

$$\theta = \theta - \beta\nabla_\theta \sum\nolimits_{T_i\sim p(T)} L_{T_i}\left(f_{\theta_i'}\right), \tag{19}$$

where T_i represents the i-th task and θ represents the updated gradient of the model following learning of a set of tasks. During the model gradient update process, the main purpose of parameter updating is to enhance the model ability to learn autonomously by learning different tasks, thereby enabling prediction of the degradation trends of the key wind-turbine gearbox components under discontinuous sample conditions.

The learning goal of the key-component degradation-trend prediction model is to minimize the loss following learning of all tasks; therefore, the objective function is formulated as

$$\min_\theta \sum_{T_i\sim p(T)} L_{T_i}\left(f_{\theta_i'}\right) = \sum_{T_i\sim p(T)} L_{T_i}\left(f_{\theta-\alpha\nabla_\theta L_{T_i}(f_\theta)}\right). \tag{20}$$

Table 2
Comparison of prediction accuracy of bear degradation trend by different models.

Model	Mean ± standard deviation (%)		
	P_e (Bearing No. 1)	P_e (Bearing No. 2)	P_e (Bearing No. 2)
RNN	10.08 ± 4.81	12.35 ± 4.73	9.06 ± 3.76
LSTM	8.51 ± 3.75	9.97 ± 5.76	8.09 ± 3.52
GRU	7.27 ± 2.29	10.25 ± 3.64	7.76 ± 3.26
Bi-LSTM	6.41 ± 2.08	7.35 ± 3.92	5.97 ± 2.27
Bi-GRU	6.62 ± 2.13	6.98 ± 2.51	5.99 ± 1.77
Transformer	6.14 ± 2.41	6.24 ± 2.27	4.27 ± 1.44
Transformer-XL	5.58 ± 2.04	5.75 ± 2.31	4.09 ± 1.27
Proposed method	4.33 ± 1.86	5.11 ± 1.64	2.78 ± 0.92

Fig. 15. Photograph of wind-turbine gearbox.

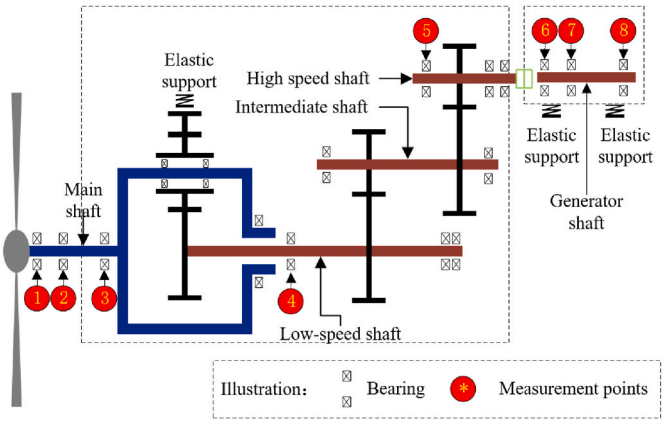


Fig. 16. Wind-turbine gearbox internal structure and CMS measurement-point location marking.

When meta-learning is applied to the degradation trend prediction of wind turbine gearboxes, the loss function $L_{T_i}\left(f_{\theta-\alpha\nabla_\theta L_{T_i}(f_\theta)}\right)$ for a single prediction task can be written as $\left\|f_\phi(\mathbf{x}^j)-\mathbf{y}^j\right\|_2^2$. Consequently, during the training of a prediction task, the loss function is defined as

$$L_{T_i}\left(f_\phi\right)=\sum_{\mathbf{x}^j,\mathbf{y}^j\sim T_i}\left\|f_\phi(\mathbf{x}^j)-\mathbf{y}^j\right\|_2^2, \tag{21}$$

where x^j represents the j-th real data and y^j represents the corresponding predicted data. In addition, to achieve a key-component degradation-trend warning, EWMA is used as the upper control threshold [27,28].

$$ULC(t)=\mu_e+K\sigma_e\sqrt{\frac{\lambda}{2-\lambda}\left[1-(1-\lambda)^{2t}\right]} \tag{22}$$

where μ_e and σ_e represent the average value and standard deviation of the compression index, respectively, and K and λ are parameters determining the size of the upper control limit (UCL). To effectively compare different degradation-trend prediction methods, the prediction accuracy for the degradation trends of key wind-turbine gearbox components is evaluated by measuring the error between the predicted and actual values [29]. Here,

$$P_e=\frac{1}{N}\sum_0^N\left(\frac{RUL_t-RUL_p}{RUL_t}\right)^2\times 100\%, \tag{23}$$

where N represents the number of predicted points for the key wind-turbine gearbox components and RUL_t and RUL_p represent the true and predicted values, respectively.

Table 3
Parameters of wind turbine gearbox.

Wind turbine	Number of stages	Gear type	Number of teeth	Number of gears	Transmission ratio
Planetary gearbox	Primary	Sun wheel	21	1	5.714
		Gear ring	99	1	
		Planetary wheel	38	3	
Parallel gearbox	Primary	Driving wheel	92	1	4.381
		Passive wheel	21	1	
	Secondary	Driving wheel	94	1	4.700
		Passive wheel	20	1	

3.2. Degradation-trend prediction process

Based on the fusion features that can characterize the degradation patterns, an NTNNet is constructed under a meta-learning framework to predict the degradation trends of the gears and bearings of a wind-turbine gearbox. The prediction process is illustrated in Fig. 2.

As shown in Fig. 2, in the proposed approach, 11 time-domain and 13 frequency-domain characteristics were extracted from the vibration signals of wind-turbine gearboxes. These characteristics were then selected based on the Mon and Tre criteria and fused using a DAE network. The fused indicators were divided into training and testing tasks, with each task further split into support and query sets. The compressed indicators were divided into tasks, each containing 15 data points. The first 10 data points of each task were used to form the first row of a 5 × 10 matrix, which was sequentially increased to create five samples for the support set. The last five data points of each task were used as the query set. During the model training process, each task was input into NTNNet, and the model parameters were updated using gradient descent, as detailed in Section 3. Finally, the trained model was used to predict the degradation trends of the key components by inputting the test data.

In this study, the proposed method was validated using a bearing fatigue-test dataset and applied to predict the degradation trends of key wind-turbine gearbox components. The experimental validation and application of the proposed method are reported in Section 4.

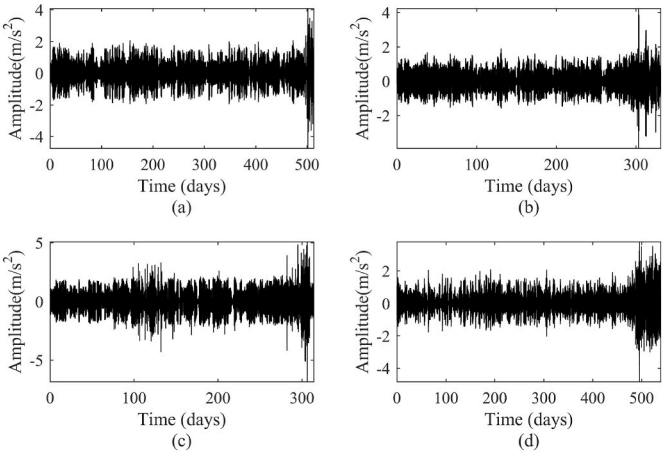


Fig. 17. Raw vibration signals of wind-turbine gearboxes: (a) bearing ball loss, (b) slight wear of inner and outer bearing rings, (c) bearing cage cracking, and (d) gear surface wear.

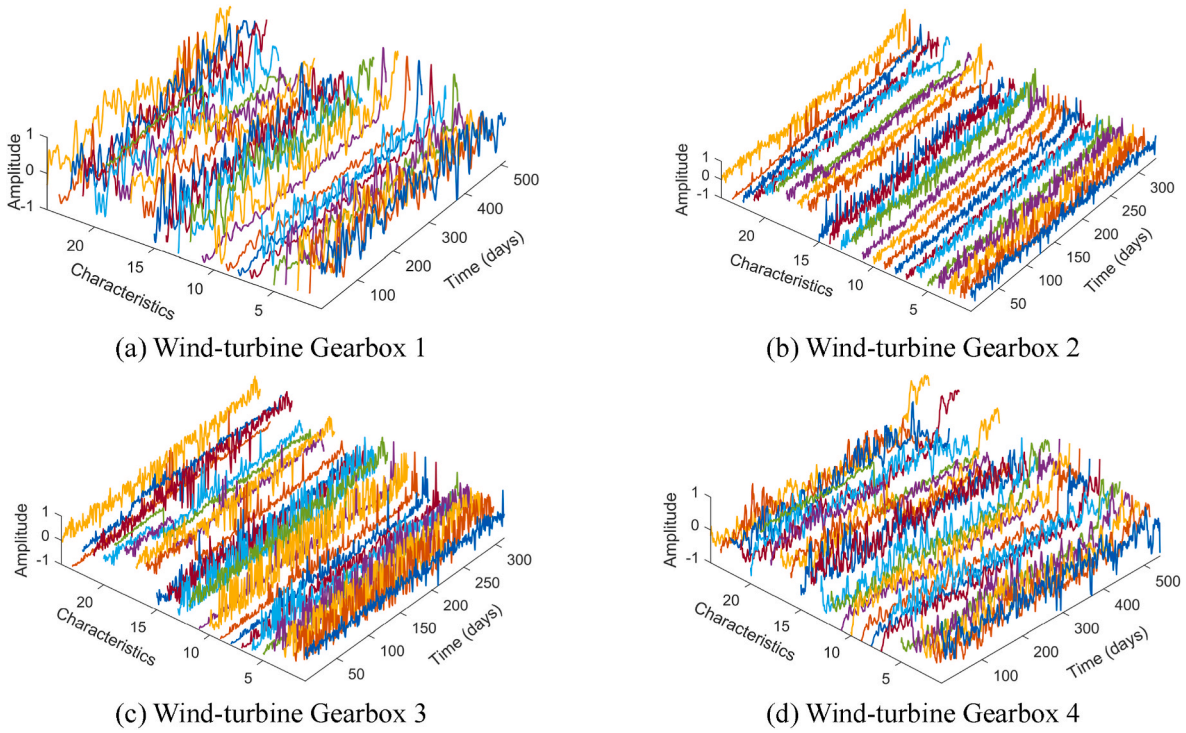


Fig. 18. Time- and frequency-domain characteristics of four wind-turbine gearbox vibration signals.

4. Experimental verification

4.1. Experimental design

To validate the accuracy of the proposed NTNNet in predicting the bearing degradation trend, fatigue tests were conducted on three bearings using a self-developed test rig. The test rig comprised a drive motor, coupling, companion bearing, test bearing, and electrical control cabinet. The vibration-signal acquisition system included two PCB 352C03 vibration sensors, an NI9234 acquisition card, an NI9188 chassis, and an acquisition software package, as shown in Fig. 3.

The tested and companion bearings were sequentially installed and a lubricant was added before the experiments were conducted. The driving motor was regulated at a constant speed of 500 rpm using an electrical control cabinet. To ensure the accurate acquisition of the bearing failure signal, the frequency of the vibration signal is set to 10,240 Hz, which is much higher than the possible fault frequency. Vibration sensors were used to capture the vibration signals of the tested bearings during the experiments. Both the tested and companion bearings were high-precision deep-groove ball bearings, with GCr15Z material comprising the inner and outer rings, as well as the rolling elements. Fig. 4 presents a photograph and dimensional drawing of the bearings. The bearing dimensional parameters are listed in Table 1.

During the experiment, test-bearing vibration signals were collected every 0.5 h, whereas those of the companion bearing were recorded every 10 h. When the companion-bearing vibration signals increased significantly or the experiment duration exceeded 200 h, the companion bearing was replaced to ensure that the test-bearing vibration signals were unaffected by the companion bearing. The experiment was terminated when the test bearing exhibited a failure frequency. Finally, the vibration signals of all three test bearings were obtained, as shown in Fig. 5.

4.2. Bearing degradation-trend prediction results

To effectively characterize the bearing degradation patterns and predict their degradation trends, time- and frequency-domain characteristics were obtained from the vibration signals of the three bearings during the fatigue testing process. The results are shown in Fig. 6.

In Fig. 6, the first 11 characteristics are time-domain indicators: the mean, standard deviation, root-mean-square value, effective value, absolute peak value, skewness, kurtosis, peak factor, margin factor, waveform factor, and impulse factor, corresponding to Indexes 0–10, respectively. The last 13 characteristics are frequency-domain indicators: the frequency-domain mean, variance, skewness, kurtosis, first-order centroid, second-order central moment, and second-order moment, along with six dimensionless frequency-domain indicators, corresponding to Indexes 18–23, respectively. As the operating time of the bearing continues to extend, the inner race, outer race, and rolling elements experience ongoing wear, leading to continuous changes in the bearing vibration signals and trend-like variations in the time-frequency domain indicators. By comprehensively considering the Mon, Tre, and Rob indicators, it is possible to effectively screen out the indicators that better characterize the bearing degradation trend, thereby achieving the prediction of the bearing degradation trend.

As shown in Fig. 6, the characteristics of the bearing vibration signals exhibited certain change trends with increasing test time. Some of the indicators had good *Mon*, whereas others had good *Tre*. The *Mon*, *Tre*, *Rob*, and *Cie* of the various bearing indicators were evaluated; the final evaluation results are shown in Fig. 7.

Considering both *Mon* and *Tre*, *Cie* was used to screen the bearing time- and frequency-domain characteristics. Finally, eight characteristics, corresponding to Indicators 1, 2, 7, 13, 15, 16, 22, and 23, were selected for fusion. These characteristics of vibration signals can reflect the degradation trend of the bearing. Screening indicators with good *Mon, Tre*, and *Cie* values for dimensionality reduction can retain as much state information as possible while eliminating irrelevant information. The selected characteristics were fused using a DAE network; the characteristic fusion results are shown in Fig. 8.

To validate the effectiveness of the integrated indicator, an ultra depth-of-field microscope was used to measure the inner and outer ring wear of the three test bearings. During the wear measurement process,

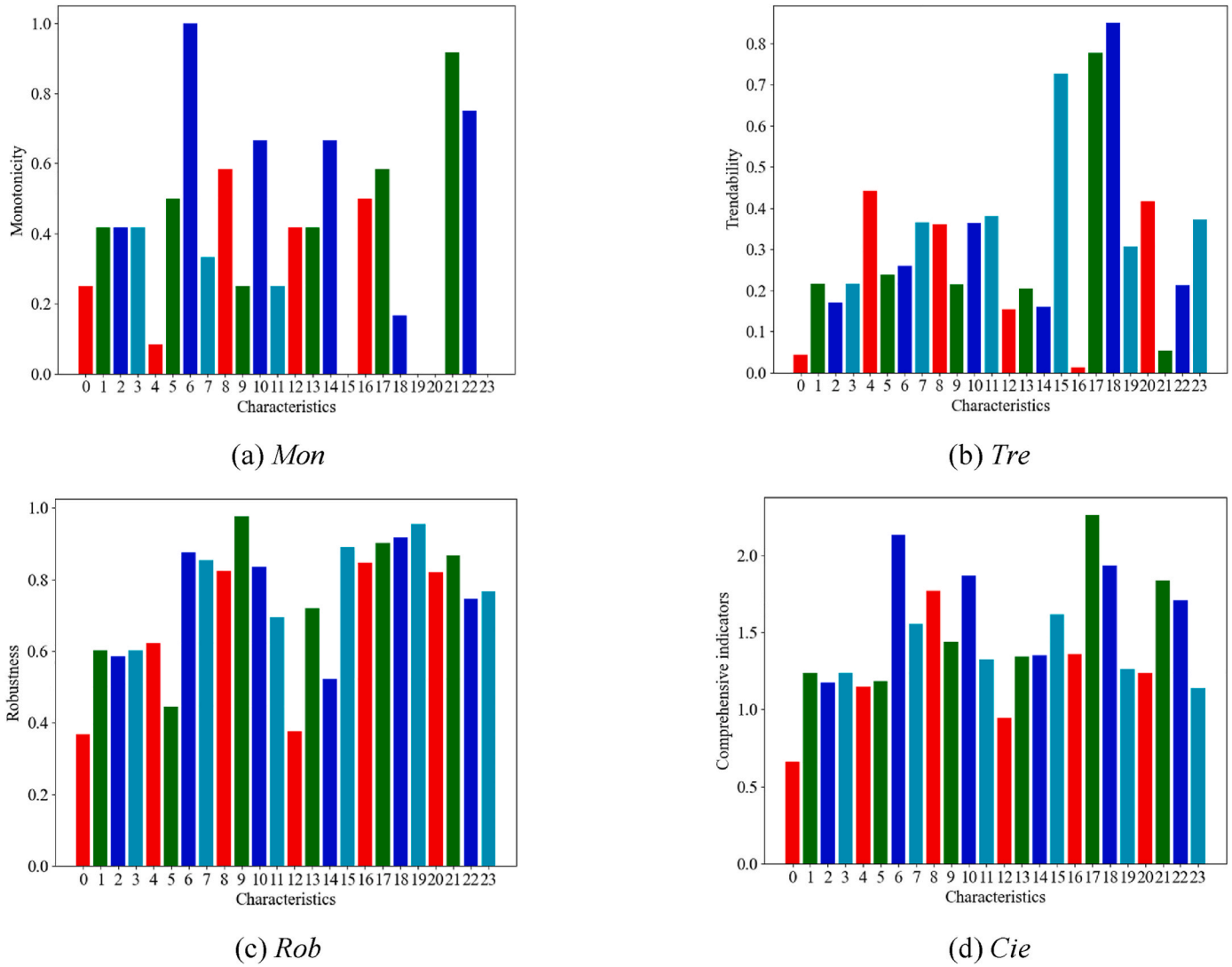


Fig. 19. Evaluation results of time- and frequency-domain characteristics of wind-turbine gearbox vibration signals.

four identical bearings were dismantled. One was not tested; the other three were subjected to fatigue tests. Photographs taken before and after the bearing dismantling are shown in Fig. 9.

To verify the effectiveness of the integrated indicator, an ultra depth-of-field microscope (VHX-700FC) was used to measure the surface wear, as shown in Fig. 10. Ultra high-pixel and ultra high-performance equipment was used, which could measure the test-piece surface morphology. This equipment could also automatically synthesize the surface 3D and measure the surface wear depth.

To quantitatively analyze the degree of bearing wear, the degrees of wear of the inner and outer bearing rings were measured. This measurement necessitated bearing disassembly; thus, it was assumed that all tested bearings had the same initial state, and a new bearing was used as the initial state for all tested bearings. The difference between the final and initial groove depths in the inner and outer bearing rings was taken as the wear value. The test results of the three tested bearings are shown in Fig. 11–13.

The distance between Points A and B in the measurement results represents the depth of the inner or outer bearing ring, and the change in depth was taken as the bearing wear degree. As the three tested bearings were of the same type, it was assumed that the initial depths of the inner and outer ring grooves of the three bearings were the same, with values of 340.4 and 311.0 μm, respectively. As shown in Fig. 11–13, at the end of the durability test, the inner-ring groove depth of Bearing 1 was 372.8 μm, with a wear depth of 32.4 μm, whereas the final outer-ring groove depth was 330.3 μm, with a wear depth of 19.3 μm. At the end of the durability test, the inner-ring groove depth of Bearing 2 was 370.6 μm, with a wear depth of 30.2 μm, and the final outer-ring groove depth was 330.1 μm, with a wear depth of 19.1 μm. At the end of the durability test, the inner-ring groove depth of Bearing 3 was 369.9 μm, with a wear depth of 29.5 μm, and the final outer-ring groove depth was 329.7 μm, with a wear depth of 18.7 μm. Based on an analysis of the inner- and outer-ring groove wear amounts of the three bearings, it can be concluded that Bearing 1 suffered the most severe wear, followed by Bearing 2 and then Bearing 3. The actual wear amounts of the three bearings corresponded to the constructed fused degradation-trend index, which demonstrates that the integrated indicator can characterize the bearing degradation pattern and can be applied to bearing degradation-trend prediction.

The integrated indicator was input into the proposed model for bearing degradation-trend prediction. In the experiments, the learning rate, dropout rate, iteration number, batch size, and decay rate of the proposed models were set to 0.001, 0.5, 200, 64, and 0.95, respectively. The activation function was relu, and the optimizer was Adam. The proposed model was run using the deep learning framework TensorFlow 2.10.0 on a system equipped with a 13th Gen Intel(R) Core(TM) i9-13900K processor, an NVIDIA GeForce RTX 4080 GPU, and 64 GB of RAM. The results obtained for the three bearings are shown in Fig. 14.

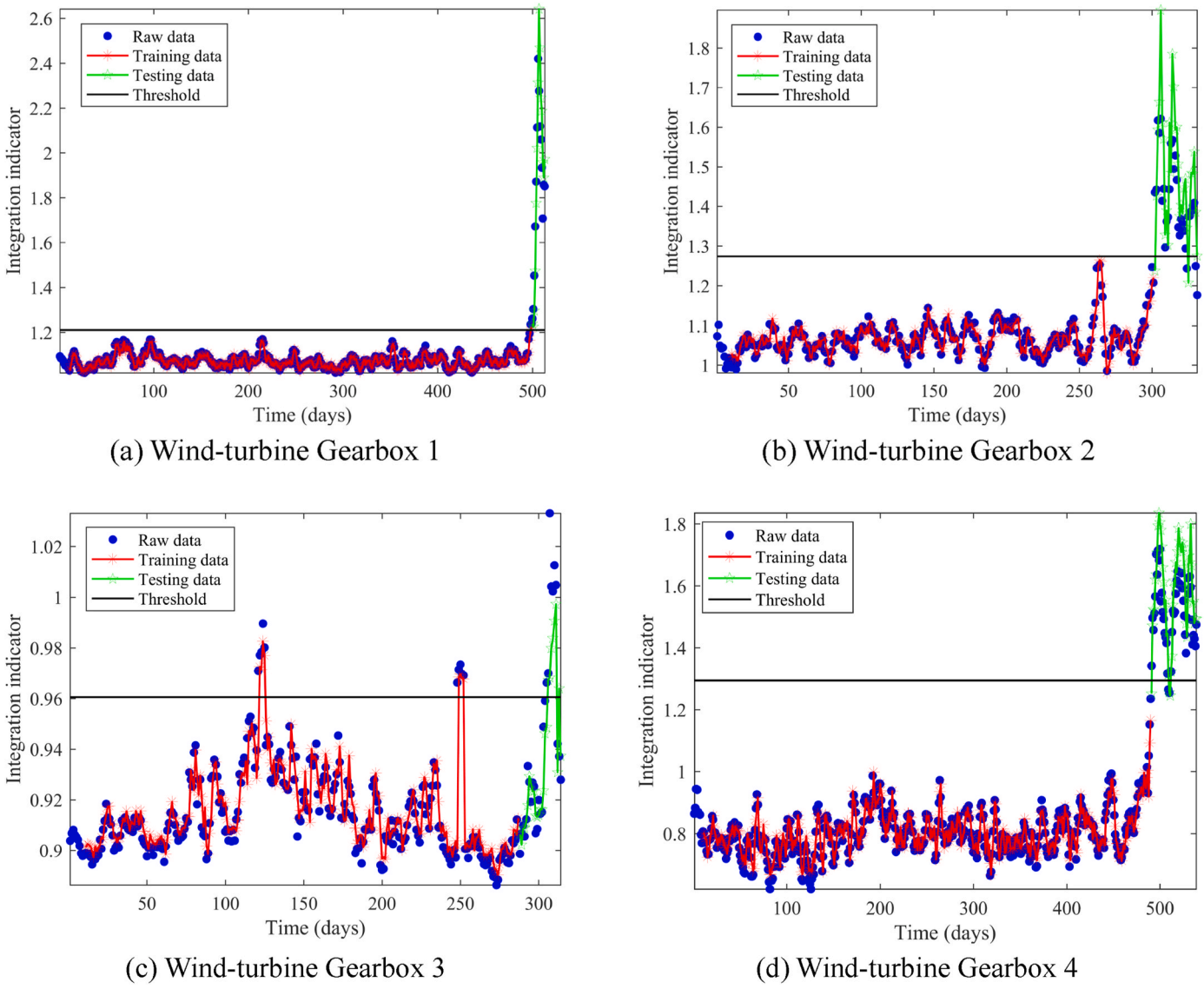


Fig. 20. Gear or bearing degradation-trend results for wind-turbine gearboxes.

Table 4
Comparison of model accuracies for prediction of degradation trends of key wind-turbine gearbox components.

Model	Mean ± standard deviation (%)			
	Gearbox No.1	Gearbox No.2	Gearbox No.3	Gearbox No.4
RNN	8.87 ± 4.05	12.51 ± 6.80	8.87 ± 4.05	9.53 ± 3.22
LSTM	8.11 ± 3.69	9.93 ± 5.71	8.11 ± 3.69	8.26 ± 3.09
GRU	6.72 ± 3.92	8.74 ± 4.47	6.72 ± 3.92	6.19 ± 1.52
Bi-LSTM	5.22 ± 2.72	6.12 ± 3.52	5.22 ± 2.72	5.51 ± 1.82
Bi-GRU	5.42 ± 2.54	5.75 ± 2.04	5.42 ± 2.54	4.18 ± 1.50
Transformer	5.14 ± 1.76	5.07 ± 1.81	5.14 ± 1.76	3.96 ± 1.64
Transformer-XL	5.21 ± 1.38	4.91 ± 1.27	5.21 ± 1.38	3.31 ± 1.43
Proposed method	4.61 ± 1.09	4.73 ± 1.34	4.61 ± 1.09	3.05 ± 1.21

Based on the results shown in Fig. 14, the proposed method integrates the indicators with clear degradation trends and can effectively learn the bearing performance evolution law. When a fault occurs, the integrated indicator increases rapidly, demonstrating that the proposed model can effectively predict bearing degradation trends. The fused indicators of the three bearings exhibit clear degradation trends. The degradation trends of Bearings 1 and 3 were better than that of Bearing 2. This phenomenon primarily occurred because of the influence of the test-bearing assembly when the companion bearings were replaced during the Bearing-2 experiment, which caused a small increase in the test-bearing vibration signal. The proposed method addresses the issue of training models with discontinuous samples by sampling short time series data, constructing multiple support and query sets, and employing multi-task learning training approaches. Additionally, a GRU is constructed within a meta-learning framework, and parameters are updated through multiple recursive iterations, thereby enhancing the representation capability of bearing degradation performance and accurately predicting the degradation trend. Overall, the integrated indicator can characterize bearing degradation laws, and the proposed model can satisfy the requirements of bearing degradation-trend prediction.

To quantitatively demonstrate the effectiveness and superiority of the proposed NTNNet in predicting bearing degradation trends, it was compared with other networks: a recurrent neural network (RNN), LSTM, GRU, Bi-LSTM, Bi-GRU, transformer [30], and transformer-XL [31] under the same conditions. Table 2 presents the results of this comparative analysis.

From Table 2, the proposed NTNNet outperformed the compared models in predicting the bearing degradation trend, achieving higher

prediction accuracies for all bearings than the RNN, LSTM, GRU, Bi-LSTM, Bi-GRU, Transformer, and Transformer-XL models. Specifically, the proposed model improved the prediction accuracy of the Bearing-1 degradation trend by 5.75 %, 4.18 %, 2.94 %, 2.08 %, 2.29 %, 1.81 %, and 1.25 % compared to the RNN, LSTM, GRU, Bi-LSTM, Bi-GRU, Transformer, and Transformer-XL models, respectively. For Bearing 2, the proposed model improved the prediction accuracy by 7.24 %, 4.86 %, 5.14 %, 2.24 %, 1.87 %, 1.13 %, and 0.64 % compared to the same models. For Bearing 3, the proposed model achieved average accuracy improvement of 6.28 %, 5.13 %, 4.98 %, 3.19 %, 3.21 %, 1.49 %, and 1.31 % compared to the same models. Additionally, the standard deviation of the proposed-model prediction accuracy was lower than those of the compared models. The main reasons for this outcome are the following. When presented with discontinuous samples, the compared models could not sufficiently learn the bearing degradation patterns and, thus, their post-training generalization performance was poor. In contrast, the proposed method utilized multitask learning to optimize the NTNNet initialization parameters with discontinuous data. This approach not only enabled the model to fully learn the bearing degradation patterns but also enhanced the model generalization performance. In summary, the proposed model exhibits improved bearing degradation-trend prediction accuracy compared to existing models.

5. Applications

The bearing durability experiments reported in the previous section demonstrate the effectiveness of the proposed method in predicting degradation trends and indicate that this method can be applied to the degradation-trend prediction of key components in wind-turbine gearboxes. Wind turbines are typically installed at high altitudes in outdoor environments, where they are subjected to severe and varying conditions such as changing wind loads and temperature differentials. These conditions significantly impact the service lifetimes of gearboxes. In particular, faults in bearings and gears exposed to alternating loads are a major cause of reduced lifetimes [32–34]. Vibration signals from wind-turbine gearboxes often exhibit discontinuities due to adverse operating environments, sporadic sensor anomalies, and unstable data transmission. Accurately predicting the degradation trends of key components based on these discontinuous samples is a significant challenge [35]. To enable precise health management of wind-turbine gearboxes, the model proposed in this study can be effectively applied to predict the degradation trends of key components, even under such challenging conditions.

5.1. Wind-turbine gearbox

The wind-turbine drivetrain consists of a primary planetary gearbox and a secondary parallel gearbox, as shown in Fig. 15. In the planetary gearbox, a planetary carrier serves as the power input and a sun gear is used as the output. In the parallel gearbox, the input shaft is rigidly connected to the planetary-box sun gear for power input. The speed is then reduced and the torque is increased through two gear stages before being transmitted to the generator via a coupling, thus completing the power transmission process.

The primary structural parameters of the wind-turbine gearbox are listed in Table 3. Existing wind turbines are equipped with a conditioning monitoring system (CMS), which mainly uses eight vibration sensors to monitor the key gearbox components [36,37]. The vibration-sensor monitoring positions are located at the front and rear bearings of the main shaft, the gear ring inside the planetary gearbox, the low-speed shaft bearing, the high-speed shaft bearing of the parallel gearbox, the generator drive-shaft bearing, the generator main-shaft drive end bearing (radial direction), the generator main-shaft drive end bearing (axial direction), and the generator main shaft free-end bearing, as shown in Fig. 16.

To validate the accuracy of the proposed model in predicting the degradation trends of key wind-turbine gearbox components, common faults were selected for experimental testing. The vibration signals from the four wind-turbine gearboxes closest to the components were used to predict the key-component degradation trends. Vibration signals from each wind turbine gearbox were collected every 4 h, with signal collection continuing until a key component failure occurred and the wind-turbine system was deactivated. Each collected signal was 128 K in length with a sampling frequency of 25,600 Hz. The vibration signals collected from the four wind-turbine gearboxes are presented in Fig. 17.

5.2. Predicted wind-turbine gearbox degradation trends

The time- and frequency-domain characteristics of the vibration signals from four wind-turbine gearboxes were extracted, and feature selection and fusion were performed. By selecting features with good *Mon* and *Tre*, integrated indicators were constructed for the key wind-turbine gearbox components. The time- and frequency-domain characteristics extracted from the vibration signals of the four wind turbine gearboxes are shown in Fig. 18.

Based on the results shown in Fig. 18, both the time- and frequency-domain characteristics of the vibration signals exhibited clear trends with extension of the wind-turbine gearbox service time. With the extension of the service life of wind turbine gearboxes, most characteristics show a gradual upward trend, such as the mean, variance, and kurtosis of vibration signals. The main reason for this phenomenon is that as the bearings and gears in the gearbox gradually wear and degrade, the vibration amplitude of each component increases, resulting in some fault frequencies. The time-frequency domain characteristics show a certain degree of monotonicity and trend changes.

To obtain a degradation-prediction indicator for the key components with positive trend characteristics, the *Mon*, *Tre*, *Rob*, and *Cie* of all characteristics were analyzed. The final evaluation results are presented in Fig. 19.

Based on the evaluation results shown in Fig. 19, some of the time- and frequency-domain characteristics of the vibration signals exhibited clear *Mon* and *Tre*. For instance, Characteristics 5, 6, 8, 10, 14, 17, 21, and 22 exhibited positive *Mon*, whereas Characteristics 4, 6, 7, 8, 11, 15, 17, and 18 exhibited good *Tre*. By considering the *Mon*, *Tre*, and *Rob* of these characteristics, eight (6, 8, 10, 15, 17, 18, 21, and 22) were selected for fusion. The proposed model was then used to predict the degradation trend of the key wind-turbine gearbox components based on the integrated indicator; the results are shown in Fig. 20.

As shown in Fig. 20, the proposed method can accurately predict the degradation trends of the key wind-turbine gearbox components. By learning from historical data for gearbox components, the proposed method can capture gear or bearing degradation patterns and achieve an accurate degradation-trend prediction. Specifically, the degradation-trend predictions for the gears and bearings in Gearboxes 1, 2, and 4 were the most accurate, whereas that for the bearing in Gearbox 3 was poor. The reason for this phenomenon is that the faults in Gearboxes 1, 2, and 4 all involved gradual deterioration of the gears and bearings, which could be better learned by the proposed model. In contrast, the fault in Gearbox 3 was a sudden crack in the bearing cage, which had strong transient characteristics and yielded a short bearing performance degradation time and a complex bearing evolution pattern. In addition, the anomaly point in Fig. 20 (c) is mainly due to the sudden increase in the gearbox speed caused by the external wind force during the actual operation of the wind turbine gearbox, resulting in a sudden increase in the constructed degradation trend index.

To quantitatively demonstrate the accuracy of the proposed NTNNet for the degradation trends of key wind-turbine gearbox components, a comparison was performed with an RNN, LSTM, GRU, Bi-LSTM, Bi-GRU, Transformer, and Transformer-XL network under the same conditions. The results are summarized in Table 4.

From Table 4, the proposed NTNNet exhibits higher prediction accuracy for the degradation trends of wind-turbine gearboxes than the

compared models. For the bearing of Wind-turbine Gearbox 1, the proposed method yielded prediction-accuracy improvements of 4.26 %, 3.50 %, 2.11 %, 0.61 %, 0.81 %, 0.53 %, and 0.60 % over the RNN, LSTM, GRU, Bi-LSTM, Bi-GRU, Transformer, and Transformer-XL models, respectively. For the bearing of Wind-turbine Gearbox 2, the proposed model achieved accuracy improvements of 7.78 %, 5.20 %, 4.01 %, 1.39 %, 1.02 %, 0.34 %, and 0.18 % over the other models, respectively. Moreover, for bearings of Wind-turbine Gearboxes 3 and 4, the proposed model outperformed the compared models by 7.18 %, 5.39 %, 4.47 %, 3.51 %, 2.78 %, 2.04 %, and 1.68 %, and by 6.48 %, 5.21 %, 3.14 %, 2.46 %, 1.13 %, 0.91 %, and 0.26 %, respectively. In addition, the standard deviations of the results predicted by the proposed model were lower than those for the compared models. The main reason for this is that a wind-turbine gearbox vibration signal has a large sampling-time interval, and the samples used to train gear or bearing degradation-trend prediction models are discontinuous. With discontinuous samples, it is difficult for conventional models to fully learn the degradation patterns of the key components in the wind-turbine gearbox. However, the proposed NTNNet utilizes a multitasking learning approach to effectively improve the initialization parameters, optimize the model generalization performance, and fully learn the degradation patterns of the wind-turbine gearbox gears and bearings from discontinuous data. Consequently, the proposed model exhibits significantly improved prediction accuracy for the degradation trends of key wind-turbine gearbox components.

6. Conclusions

This study proposed NTNNet, a method involving a multitask learning approach to improve the prediction accuracy of the degradation trends of key wind-turbine gearbox components. Existing models exhibit poor generalization owing to insufficient sample continuity. The proposed method adequately extracts sample features and adaptively adjusts the model initialization parameters under the condition of sample discontinuity, thereby improving the prediction accuracy for the key-component degradation trends. Key-component integrated indicators are constructed by extracting, compressing, and fusing the characteristics of the wind-turbine gearbox vibration signals. The proposed NTNNet is then used to predict the degradation trends of key wind-turbine gearbox components. In this study, the model was validated by fatigue tests for three pairs of gears and was successfully applied to predict the degradation trends of key wind-turbine gearbox components.

The main conclusions of this study are as follows. First, the proposed method effectively constructs integrated indicators for key wind-turbine gearbox components, such as gears and bearings, by extracting, selecting, and fusing the characteristics. Second, the proposed method enhances the representation ability of NTNNet for the degradation mechanisms of key wind-turbine gearbox components, enabling it to learn the degradation patterns of gears and bearings under the condition of sample discontinuity and, thus, to exhibit improved degradation-trend prediction accuracy. Third, in an experiment, the proposed model outperformed RNN, LSTM, GRU, Bi-LSTM, Bi-GRU, Transformer, and Transformer-XL models under the same conditions when predicting the degradation trends of key wind-turbine gearbox components.

In summary, the proposed method can predict gear or bearing degradation trends, satisfy the degradation-trend prediction requirements for these components, and be effectively utilized in wind-turbine gearbox health assessments. In future research, physical mechanisms will be added to the network to further improve the accuracy of the wind gearbox degradation-trend prediction.

CRediT authorship contribution statement

Xiaoxia Yu: Methodology, Funding acquisition. **Zhigang Zhang:** Formal analysis. **Baoping Tang:** Data curation. **Jinghua Ma:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] X. Li, W. Teng, D. Peng, T. Ma, X. Wu, Y. Liu, Feature fusion model based health indicator construction and self-constraint state-space estimator for remaining useful life prediction of bearings in wind turbines, Reliab. Eng. Syst. Saf. 233 (2023) 109124, https://doi.org/10.1016/j.ress.2023.109124.
[2] I. de Pater, M. Mitici, Developing health indicators and RUL prognostics for systems with few failure instances and varying operating conditions using a LSTM autoencoder, Eng. Appl. Artif. Intell. 117 (2023) 105582, https://doi.org/10.1016/j.engappai.2022.105582.
[3] Y. Qin, J. Yang, J. Zhou, H. Pu, Y. Mao, A new supervised multi-head self-attention autoencoder for health indicator construction and similarity-based machinery RUL prediction, Adv. Eng. Inf. 56 (2023) 101973, https://doi.org/10.1016/j.aei.2023.101973.
[4] Y. Zhu, C. Zhu, J. Tan, Y. Tan, L. Rao, Anomaly detection and condition monitoring of wind turbine gearbox based on LSTM-FS and transfer learning, Renew. Energy 189 (2022) 90–103, https://doi.org/10.1016/j.renene.2022.02.061.
[5] K. He, Z. Su, X. Tian, H. Yu, M. Luo, RUL prediction of wind turbine gearbox bearings based on self-calibration temporal convolutional network, IEEE Trans. Instrum. Meas. 71 (2022), https://doi.org/10.1109/TIM.2022.3143881.
[6] S.V.V.S.N. Pichika, R. Yadav, S. Geetha Rajasekharan, H.M. Praveen, V. Inturi, Optimal sensor placement for identifying multi-component failures in a wind turbine gearbox using integrated condition monitoring scheme, Appl. Acoust. 187 (2022) 108505, https://doi.org/10.1016/j.apacoust.2021.108505.
[7] B. Merainani, S. Laddada, E. Bechhoefer, M.A.A. Chikh, D. Benazzouz, An integrated methodology for estimating the remaining useful life of high-speed wind turbine shaft bearings with limited samples, Renew. Energy 182 (2022) 1141–1151, https://doi.org/10.1016/j.renene.2021.10.062.
[8] N. Li, P. Xu, Y. Lei, X. Cai, D. Kong, A self-data-driven method for remaining useful life prediction of wind turbines considering continuously varying speeds, Mech. Syst. Signal Process. 165 (2022) 108315, https://doi.org/10.1016/j.ymssp.2021.108315.
[9] Z. Gao, P. Odgaard, Real-time monitoring, fault prediction and health management for offshore wind turbine systems, Renew. Energy 218 (2023), https://doi.org/10.1016/j.renene.2023.119258.
[10] W. Jiang, J. Wu, H. Zhu, X. Li, L. Gao, Paired ensemble and group knowledge measurement for health evaluation of wind turbine gearbox under compound fault scenarios, J. Manuf. Syst. 70 (2023) 382–394, https://doi.org/10.1016/j.jmsy.2023.08.004.
[11] X. Li, W. Teng, D. Peng, T. Ma, X. Wu, Y. Liu, Feature fusion model based health indicator construction and self-constraint state-space estimator for remaining useful life prediction of bearings in wind turbines, Reliab. Eng. Syst. Saf. 233 (2023) 109124, https://doi.org/10.1016/j.ress.2023.109124.
[12] M. Rezamand, M. Kordestani, R. Carriveau, D.S.K. Ting, M.E. Orchard, M. Saif, Critical wind turbine components prognostics: a comprehensive review, IEEE Trans. Instrum. Meas. 69 (2020) 9306–9328, https://doi.org/10.1109/TIM.2020.3030165.
[13] S. Xiang, Y. Qin, J. Luo, F. Wu, K. Gryllias, A concise self-adapting deep learning network for machine remaining useful life prediction, Mech. Syst. Signal Process. 191 (2023) 110187, https://doi.org/10.1016/j.ymssp.2023.110187.
[14] R. Bai, K. Noman, K. Feng, Z. Peng, Y. Li, A two-phase-based deep neural network for simultaneous health monitoring and prediction of rolling bearings, Reliab. Eng. Syst. Saf. 238 (2023) 109428, https://doi.org/10.1016/j.ress.2023.109428.
[15] L. Cao, H. Zhang, Z. Meng, X. Wang, A parallel GRU with dual-stage attention mechanism model integrating uncertainty quantification for probabilistic RUL

prediction of wind turbine bearings, Reliab. Eng. Syst. Saf. 235 (2023) 109197, https://doi.org/10.1016/j.ress.2023.109197.
[16] H. Yan, Y. Qin, S. Xiang, Y. Wang, H. Chen, Long-term gear life prediction based on ordered neurons LSTM neural networks, Meas. J. Int. Meas. Confed. 165 (2020) 108205, https://doi.org/10.1016/j.measurement.2020.108205.
[17] Y. Yu, C. Hu, X. Si, J. Zheng, J. Zhang, Averaged Bi-LSTM networks for RUL prognostics with non-life-cycle labeled dataset, Neurocomputing 402 (2020) 134–147, https://doi.org/10.1016/j.neucom.2020.03.041.
[18] Y. Zhao, G. Zhang, W. Hu, Q. Huang, Z. Chen, F. Blaabjerg, Meta-learning based voltage control strategy for emergency faults of active distribution networks, Appl. Energy 349 (2023) 121399, https://doi.org/10.1016/j.apenergy.2023.121399.
[19] P.W. Khan, C.Y. Yeun, Y.C. Byun, Fault detection of wind turbines using SCADA data and genetic algorithm-based ensemble learning, Eng. Fail. Anal. 148 (2023), https://doi.org/10.1016/j.engfailanal.2023.107209.
[20] L. Qiao, Y. Zhang, Q. Wang, Fault detection in wind turbine generators using a meta-learning-based convolutional neural network, Mech. Syst. Signal Process. 200 (2023) 110528, https://doi.org/10.1016/j.ymssp.2023.110528.
[21] J. Yang, X. Wang, Z. Luo, Few-shot remaining useful life prediction based on meta-learning with deep sparse kernel network, Inf. Sci. 653 (2024) 119795, https://doi.org/10.1016/j.ins.2023.119795.
[22] J. Yang, X. Wang, Meta-learning with deep flow kernel network for few shot cross-domain remaining useful life prediction, Reliab. Eng. Syst. Saf. 244 (2024) 109928, https://doi.org/10.1016/j.ress.2024.109928.
[23] T. Pan, J. Chen, Z. Ye, A. Li, A multi-head attention network with adaptive meta-transfer learning for RUL prediction of rocket engines, Reliab. Eng. Syst. Saf. 225 (2022) 108610, https://doi.org/10.1016/j.ress.2022.108610.
[24] Z. Zhao, Y. Jiao, Y. Xu, Z. Chen, E. Zio, A fault diagnosis framework using unlabeled data based on automatic clustering with meta-learning, Eng. Appl. Artif. Intell. 139 (2025), https://doi.org/10.1016/j.engappai.2024.109584.
[25] S. Xiang, Y. Qin, C. Zhu, Y. Wang, H. Chen, LSTM networks based on attention ordered neurons for gear remaining life prediction, ISA Trans. 106 (2020) 343–354, https://doi.org/10.1016/j.isatra.2020.06.023.
[26] K. Javed, R. Gouriveau, N. Zerhouni, P. Nectoux, K. Javed, R. Gouriveau, N. Zerhouni, P. Nectoux, E. Health, K. Javed, R. Gouriveau, N. Zerhouni, P. Nectoux, Enabling health monitoring approach based on vibration data for accurate prognostics. To Cite This Version : HAL Id : Hal-01032080 Enabling Health Monitoring Approach Based on Vibration Data for Accurate Prognostics, 2014.
[27] D. Chen, W. Cai, H. Yu, F. Wu, Y. Qin, A novel transfer gear life prediction method by the cross-condition health indicator and nested hierarchical binary-valued network, Reliab. Eng. Syst. Saf. 237 (2023) 109390, https://doi.org/10.1016/j.ress.2023.109390.
[28] Y. Qin, J. Yang, J. Zhou, H. Pu, Y. Mao, A new supervised multi-head self-attention autoencoder for health indicator construction and similarity-based machinery RUL prediction, Adv. Eng. Inf. 56 (2023) 101973, https://doi.org/10.1016/j.aei.2023.101973.
[29] L. Guo, N. Li, F. Jia, Y. Lei, J. Lin, A recurrent neural network based health indicator for remaining useful life prediction of bearings, Neurocomputing 240 (2017) 98–109, https://doi.org/10.1016/j.neucom.2017.02.045.
[30] J.U. Ashish Vaswani, Noam Shazeer, Niki Parmar, attention is all you need, IEEE Ind. Appl. Mag. 8 (2017) 8–15, https://doi.org/10.1109/2943.974352.
[31] Z. Dai, Z. Yang, Y. Yang, J. Carbonell, Q.V. Le, R. Salakhutdinov, Transformer-XL: attentive language models beyond a fixed-length context, ACL 2019 - 57th Annu. Meet, Assoc. Comput. Linguist. Proc. Conf. (2020) 2978–2988, https://doi.org/10.18653/v1/p19-1285.
[32] T. Adedipe, M. Shafiee, E. Zio, Bayesian network modelling for the wind energy industry: an overview, Reliab. Eng. Syst. Saf. 202 (2020) 107053, https://doi.org/10.1016/j.ress.2020.107053.
[33] M. Rezamand, M. Kordestani, M.E. Orchard, R. Carriveau, D.S.K. Ting, M. Saif, Improved remaining useful life estimation of wind turbine drivetrain bearings under varying operating conditions, IEEE Trans. Ind. Inf. 17 (2021) 1742–1752, https://doi.org/10.1109/TII.2020.2993074.
[34] F.K. Moghadam, A.R. Nejad, Online condition monitoring of floating wind turbines drivetrain by means of digital twin, Mech. Syst. Signal Process. 162 (2022) 108087, https://doi.org/10.1016/j.ymssp.2021.108087.
[35] X. Yu, B. Tang, L. Deng, Fault diagnosis of rotating machinery based on graph weighted reinforcement networks under small samples and strong noise, Mech. Syst. Signal Process. 186 (2023) 109848, https://doi.org/10.1016/j.ymssp.2022.109848.
[36] Y. Nie, F. Li, L. Wang, J. Li, M. Sun, M. Wang, J. Li, A mathematical model of vibration signal for multistage wind turbine gearboxes with transmission path effect analysis, Mech. Mach. Theor. 167 (2022) 104428, https://doi.org/10.1016/j.mechmachtheory.2021.104428.
[37] K. Zhang, B. Tang, L. Deng, Q. Tan, H. Yu, A fault diagnosis method for wind turbines gearbox based on adaptive loss weighted meta-ResNet under noisy labels, Mech. Syst. Signal Process. 161 (2021) 107963, https://doi.org/10.1016/j.ymssp.2021.107963.

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